Shutong Zheng

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Google Scholar | github.com/Yodeesy

Research Interests: Reinforcement Learning (Theory & Robustness), Federated Learning, Multi-Agent Systems

Education

Sun Yat-sen University (SYSU) Guangzhou, China
B.S. in Information and Computing Science, School of Computer Science and EngineeringSep. 2023 – Jun. 2027 (Expected)

- **Academic Standing:** Cumulative GPA: 3.0/4.0 | Major Rank: 28/45
- **Selected High-Scoring Courses:** Artificial Intelligence (**100**/100), Parallel Computer Architecture (**95**/100), High-Performance Computing (**94**/100), Information Security (**93**/100), Neural Networks (**90**/100)

Publications & Working Papers

[ICLR 2027 Under Review] When to Intervene? State-Aware Sparse Manipulation in Federated Reinforcement Learning *First Author*

- Formalized intervention timing as a novel vulnerability dimension in FRL; developed V-BSA using policy-entropy gating and envelope-constrained behavioral steering to theoretically and empirically expose limits of robust aggregation rules.

[AAAI 2027 Under Review] Rethinking One-Shot Federated Graph Learning: Training-Free Statistical Estimation *First Author*

- Identified representation misalignment from local GNN training under extreme non-IID graphs; proposed SPEAR, a training-free framework computing topology-smoothed prototypes with adaptive shrinkage aggregation, yielding multi-magnitude speedups.

[WSDM 2027 Under Review] Structural Entropy-Driven Graph Diffusion Generation for One-Shot Federated Graph Learning *Co-first Author (1st)*

- Resolved the volume–structure mismatch in one-shot FGL via structural entropy-based client differentiation; developed SPIRE, synthesizing topologically regularized pseudographs via server-side graph diffusion.

Research Experience

The Hong Kong University of Science and Technology (Guangzhou) Guangzhou, China
Research Assistant, AgenticFin Lab (Advisor: Prof. Sijia Chen) *Jul. 2026 – Present*

- Focus on robustness and policy stability in federated reinforcement learning (FRL) under imperfect environments, with an active extension toward agentic RL and cooperative decision-making.

Sun Yat-sen University (SYSU) Guangzhou, China
Undergraduate Researcher, InPlusLab (Advisor: Prof. Chuan Chen) *Nov. 2024 – Jun. 2026*

- Focused on federated learning and graph neural networks, primarily investigating optimization paradigms for communication-efficient one-shot federated graph learning (One-Shot FGL).

Open-Source Projects

Yquoter | High-Performance Financial Data Infrastructure GitHub: Yquoter/Yquoter
Core Maintainer & Architecture Lead *Aug. 2025 – Present*

- Engineered an asynchronous, fault-tolerant financial data fetcher in Python `asyncio` supporting high-concurrency multi-market feeds and tiered caching.

Technical Skills

Core Theory & Methods: Theoretical RL, Robust Optimization, Federated Learning, Graph ML
Languages & Tools: Python, C++, CUDA (Basic), Linux/WSL, Parallel Computing (OpenMP/MPI), Git, LaTeX
Frameworks & Libraries: PyTorch, PyTorch Geometric, HuggingFace, NumPy, SciPy